

If LLMs Misbehave, Will We Even Be Able to Tell?

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Abstract

Training large language models to perform latent reasoning may improve performance, but the lack of legible reasoning traces challenges model interpretability and safety assurance. This paper investigates whether existing interpretability methods can be used to effectively decode latent reasoning into a legible form. We find that the logit lens can substantially improve the legibility of COCONUT’s latent reasoning tokens, but explicit reasoning is still significantly more legible.

1 Introduction

Large language models (LLMs) have become increasingly central to AI research as their capabilities have been improved by scale [13]. These capabilities are especially pronounced in reasoning models, such as OpenAI o1 [12] and DeepSeek R1 [10], that learn to produce higher quality outputs by starting their responses with a chain-of-thought (CoT): a passage of text in which the model’s given task is broken down into smaller steps that can be tackled one at a time [21]. Latent reasoning models (LRMs) replace these human-readable steps with an illegible array of numbers, as shown in Figure 1. This modification may improve reasoning performance, but it also raises new challenges for our understanding of these systems.

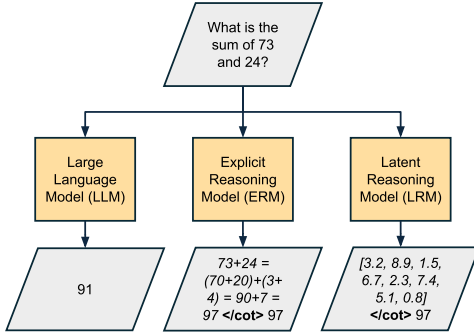


Figure 1: A comparison of model outputs.

Interpreting LLMs is difficult due to complications like polysemaniticity and superposition [5, 18]. However, explicit reasoning models have provided an unexpected advantage: the CoT that they generate can provide observers with information about model behaviour. Safeguards that leverage this information can even be much more effective than ones that don’t [1]. Korbak et al. [16] warn that we could lose the advantages conferred by an explicit CoT in the transition to LRMs.

This paper aims to address the following question: *how effective are existing interpretability methods at decoding latent reasoning into*

a legible form? Our contributions are as follows: first, in Section 3 we apply a scalable method that provides a quantitative assessment of legibility; second, in Section 4 we present evidence suggestive of a trend between observer scale and reasoning legibility.

2 Background

Latent Reasoning Models: Two distinct categories of latent reasoning models can be identified: vertical recurrence models, such as Huginn-0125 [7] and Ouro [23], that repeatedly apply intermediate layers in a deep neural network to the final hidden state; and horizontal recurrence models, such as COCONUT [11] and CODI [20], that generate latent reasoning through an autoregressive process where the final hidden state is combined with the prompt and used as input to another forward pass. In this paper we focus on COCONUT [11] which is a training procedure that produces latent reasoning models by applying a training curriculum to a base model. First, the model is fine-tuned on a dataset of explicit reasoning traces. Then, an iterative approach is used to replace one explicit reasoning step at a time with latent reasoning [11].

Latent Reasoning Interpretability: Some existing interpretability work on latent reasoning models has attempted to decode latent reasoning tokens into a legible representation. Shen et al. [20] decode latent reasoning tokens by applying the logit lens alongside attention analysis methods. The decoded latent reasoning tokens reveal a surprising pattern, which Goyal et al. [9] explained as an alternation between stored intermediate results, and transitional states produced during computations over those stored results. Dilgren and Wiegrefe [4] apply similar methods when decoding latent reasoning tokens generated by COCONUT and CODI. They find that latent reasoning tokens can be used to reconstruct gold reasoning traces. In contrast, we investigate whether latent reasoning traces are informative when gold reasoning traces are unavailable.

3 Method

We evaluate the legibility of reasoning representations with a method adapted from Kirchner et al. [15]. We sample problem statements from a dataset, and then use a reasoning model to generate responses to them. Each response consists of a sequence of reasoning tokens, and a final answer. An interpretability method is used to generate an explanation of the reasoning tokens. This explanation, alongside the sampled problem statement, is given to the observer, who is asked to predict the correctness of the reasoning model’s answer. We visualise this methodology in Figure 2. In Section 4 we report the balanced accuracy of the observer’s predictions, and in the rest of this section we explain our experimental setting¹.

¹Code available at <https://github.com/Aauustiin/Decoding-Latent-Reasoning>.

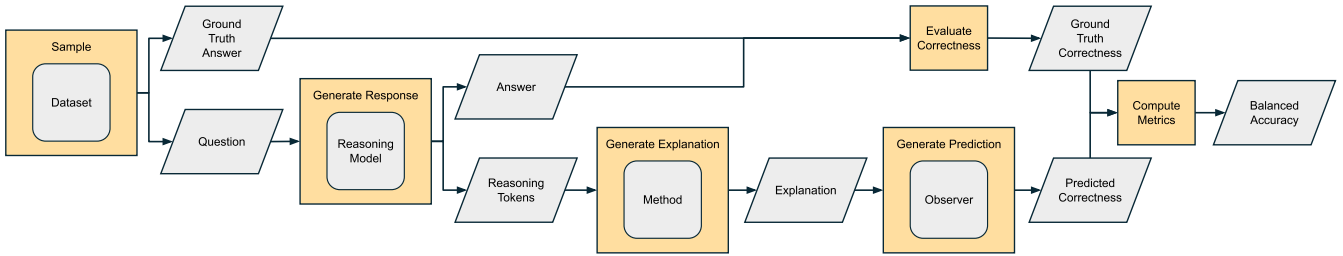


Figure 2: The method used to evaluate the legibility of reasoning representations. Rectangular boxes are processes, parallelograms are data, and rounded boxes are variables that can change between runs.

Dataset: We use the GSM8k dataset, which contains math questions solvable by a "bright middle school student" [2]. Each sample is comprised of two elements: a question, and an answer. The answer consists of workings in natural language, interspersed with key steps presented in calculator notation, followed by a delimiter, and the ground truth answer. We split this dataset into two partitions: a split comprised of 7473 samples for the training of our reasoning models, and an evaluation split comprised of 1319 samples.

Reasoning Models: We use two different reasoning models when generating responses to GSM8k questions. The first is an explicit reasoning model, produced by applying the first stage of the COCONUT training procedure [11] to GPT-2 small [19]. This stage of training requires a dataset of gold reasoning traces, which is generated by extracting the key steps in calculator notation, the delimiter, and the ground truth answer from each GSM8k sample in the training split. The second model has latent reasoning capabilities, and it is created by applying the second stage of the COCONUT training procedure [11] to our explicit reasoning model. We use the same hyperparameters as Hao et al. [11] during the training and evaluation of both models. Our ERM scores 46.3% on the GSM8k evaluation split, and our LRM scores 38.1%. These results are similar to those reported by Hao et al. [11], which were 42.9% for the ERM, and 34.1% for the LRM.

Interpretability Methods: We apply one interpretability method to our LRM: the logit lens, first presented by nostalgebraist [17]. Layer norm is applied to each latent reasoning token, and each result is then multiplied by the unembedding matrix to produce logits. We apply some post-processing to the output of the logit lens when rendering it to the observer. The softmax operation is used to normalise the logits, then the five token-score pairs with the highest scores are shown to the observer. We also test a no-explanation baseline, on both the LRM and the ERM, where all reasoning tokens are shown directly to the observer.

Observers: We test 3 observers, each of which is itself an explicit reasoning model. These three observers—Qwen3-1.7B, Qwen3-8B, and Qwen3-14B—are all members of the Qwen3 family presented by Yang et al. [22].

4 Results

Our results, presented in Figure 3, show that the predictive accuracy of observers approaches 100% when they’re given explicit reasoning,

and is at the level of random chance when they’re given latent reasoning. This result is unsurprising, and it is demonstrative of the concern raised by Korbak et al. [16].

We find that the legibility of decoded latent reasoning tokens varies with the size of the observer. When provided with the Decoded Latent reasoning tokens, Qwen3-1.7B’s predictive accuracy is at the level of random chance, whereas Qwen3-14B’s balanced accuracy rises to 65.5%, recovering 36.5% of the gap between its performance on raw latent reasoning tokens, and explicit reasoning.

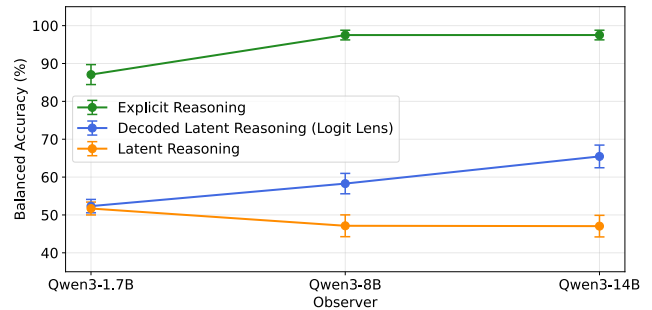


Figure 3: Balanced accuracy, with 95% confidence intervals, by observer and representation.

5 Conclusion

Our preliminary findings show that interpretability methods can substantially improve the legibility of latent reasoning, but a large gap between the legibility of latent reasoning and explicit reasoning remains. In future work we will determine whether other interpretability methods—such as sparse autoencoders [3], patchscopes [8], activation oracles [14], and natural language autoencoders [6]—can close this gap. Furthermore, our results are suggestive of a trend between observer scale and legibility, and in future work we will determine whether this trend continues when larger observers are tested.

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